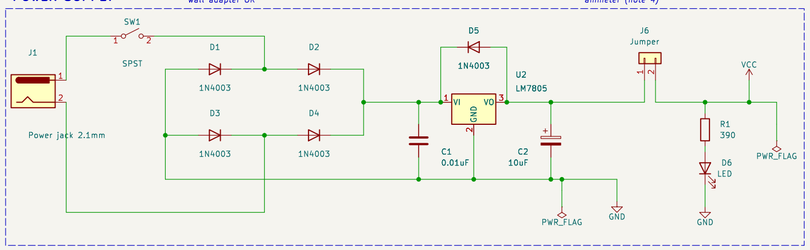
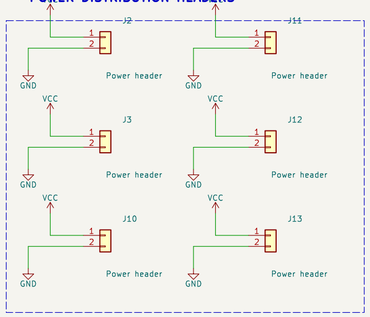
POWER SUPPLY

either polarity wall adapter OK

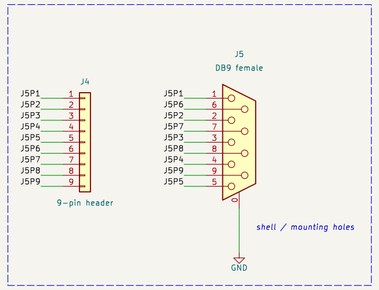
remove jumper for ammeter (note 4)



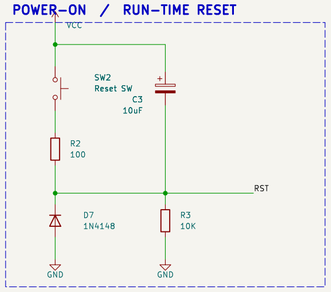
POWER DISTRIBUTION HEADERS



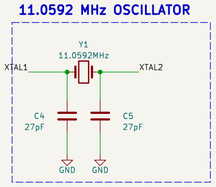
RS-232 SERIAL CONNECTOR



POWER-ON / RUN-TIME RESET



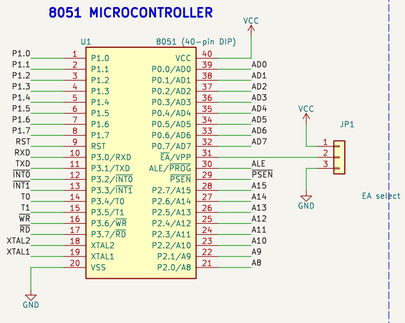
11.0592 MHz OSCILLATOR



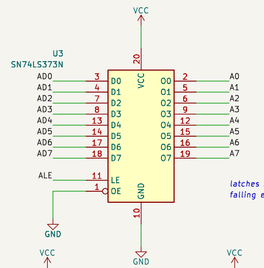


decoupling at U1 pins 20 & 40

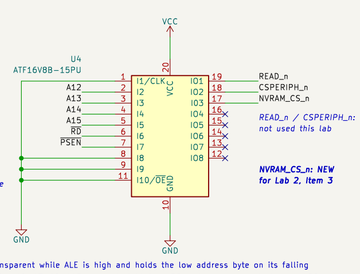
8051 MICROCONTROLLER



EXTERNAL BUS: ADDRESS LATCH AND DECODE



latches AD0–AD7 on the falling edge of ALE



READ_n / CSPERIPH_n: not used this lab

NVRAM_CS_n: NEW for Lab 2, item 3

U3 is transparent while ALE is high and holds the low address byte on its falling edge, giving A0–A7. A8–A15 come straight from P2 and need no latch, because P2 holds the high address byte for the whole bus cycle.

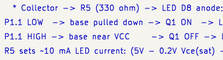
U4 (SPLD) equations — must match the pin numbers declared in the WinCUPL source (lab2.pld). NVRAM_CS_n is NEW for Lab 2 — drives the NVRAM's /CE pin:

```
READ_n      =  RD_n & PSEN_n              (Lab 1, not used this lab)
CSPERIPH_n  =  !(A15 & A14 & A13 & A12)   (Lab 1, not used this lab)
NVRAM_CS_n  =  A15 # PSEN_n               low only when fetching (PSEN low)
                                          AND address is in 0000h–7FFFh (A15=0)
```

Free-run test: jumper JP1 so /EA is low and leave the memory sockets empty. PO then floats to FFh, the CPU free-runs through the whole 64K code space and every address line becomes a binary counter. At 11.0592 MHz one machine cycle is 1.085 us, so expect ALE = 1.843 MHz, A0 = 460.8 kHz, A15 = 14.06 Hz, and CSPERIPH_n low for 4.44 ms out of every 71.1 ms sweep — a 15/16 high duty cycle.

C7 at U3, C8 at U4: 0.1 uF right at each IC's Vcc/GND pins

MOUNTING HARDWARE



H1 – H4: #4-40 x 1/2 in standoffs, 8 x #4-40 screws, one at each corner of the PCB

NOTES

1. Power input J1: 9–12 VDC unregulated wall adapter, 500 mA minimum, 2.1 mm coaxial plug. The diode bridge D1–D4 makes the input polarity insensitive, so either a center-positive or a center-negative adapter may be used; two diode drops (~1.4 V) are lost in the bridge.
2. U2 (7805 / LM340T-5.0) requires Vin > 7.5 V to stay in regulation. Fit a heat sink if the board draws more than ~250 mA.
3. D5 protects U2: if the input is shorted or collapses faster than the output, C2 discharges through D5 instead of backwards through U2.
4. J6 is normally shorted with a jumper (or a soldered wire). Remove it and insert an ammeter to measure the total 5 V board current.
5. Power-on reset: C3 = 10 uF charges through R3 = 10 k. RST is held above VIH for about R3*C*ln(Vcc/VIH) = 0.1 s * ln(5/2) = 92 ms, far more than the 2 machine cycles (2.2 us at 11.0592 MHz) the 8051 requires. SW2 gives a run-time reset; R2 = 100 ohm limits the capacitor discharge current to 50 mA. D7 clamps the negative excursion on RST when power is removed and C3 discharges.
6. 11.0592 MHz is used because it divides down to exact standard baud rates (300 – 57600 baud) with timer 1 in mode 2.
   C4 = C5 = 27 pF for a crystal load capacitance of 18 pF: CL = (C4*C5)/(C4+C5) + Cstray = 13.5 pF + ~5 pF.
7. Power-on LED current: I = (5 V – 2.0 V) / 390 ohm = 7.7 mA.
8. JP1 selects the memory mode: /EA to Vcc runs code from on-chip ROM/flash; /EA to GND forces every code fetch to external memory.
9. J4 is a 9-pin header wired 1:1 to the DB9 connector J5. An RS-232 level translator (MAX232 or equivalent) is inserted between the 8051 TXD / RXD pins and J4 (added in Lab 2, not yet on this sheet pending final RS-232 hardware). Pin 9 (RI) is not used in this course.
10. C6 (0.1 uF) is mounted directly across U1 pins 20 and 40. U1 is fitted in a 40-pin machine-tooled DIP socket.
11. Port pins P0, P2, /RD, /WR, ALE and /PSEN are labelled for the address/data bus, address latch and external memory.
12. U3 and U4 sit in 20-pin wire-wrap DIP sockets; U5 (NVRAM) sits in a 28-pin wire-wrap DIP socket. Mechanical and hardware items (PCB, sockets, standoffs, screws, power jack, switches, DB9 and jumpers) are itemised in the BOM file issued with this schematic.
13. NEW for Lab 2: U5 (NVRAM/EPROM substitute, item 3) and Q1/R4/R5/D8 (LED driver, item 4). See the dedicated boxes for wiring detail.

U3 SN74LS373N octal latch

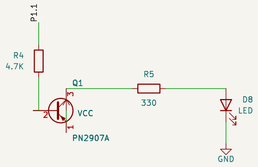
1	/OE	GND	11	LE	ALE
2	O0	A0	12	O4	A4
3	D0	AD0	13	D4	AD4
4	D1	AD1	14	D5	AD5
5	O1	A1	15	O5	A5
6	O2	A2	16	O6	A6
7	D2	AD2	17	D6	AD6
8	D3	AD3	18	D7	AD7
9	O3	A3	19	O7	A7
10	GND	GND	20	Vcc	+5V

U4 ATF16V8B-15PU SPLD

1	I1	GND spare	11	I10	GND spare
2	I2	A12	12	IO8	not used
3	I3	A13	13	IO7	not used
4	I4	A14	14	IO6	not used
5	I5	A15	15	IO5	not used
6	I6	/RD	16	IO4	not used
7	I7	/PSEN	17	IO3	NVRAM_CS_n
8	I8	GND spare	18	IO2	CSPERIPH_n
9	I9	GND spare	19	IO1	READ_n
10	GND	GND	20	Vcc	+5V

These pin assignments also serve as the wire-wrap ID labels.

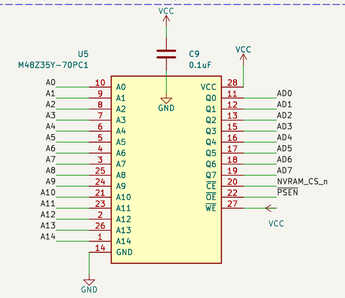
LED DRIVER (ITEM 4)



Q1 (PN2907A, PNP) is a HIGH-SIDE switch:
- Emitter –> VCC
- Base –> R4 (4.7k) –> P1.1
- Collector –> R5 (330 ohm) –> LED D8 anode; D8 cathode –> GND

P1.1 LOW –> base pulled down –> Q1 ON –> LED lit
P1.1 HIGH –> base near VCC –> Q1 OFF –> LED dark
R5 sets ~10 mA LED current: (5V – 0.2V Vce(sat) – 2.0V LED Vf) / 10mA = 280R, use standard 330R.
R4 forces hard saturation: (5V – 0.6V Vbe – 0.4V pin Vol) / 4.7k = 0.85 mA base current, well above Ic/hFE for PN2907A at 10 mA.

EXTERNAL MEMORY: NVRAM (EPROM SUBSTITUTE)



U5 is the physical BQ4011YMA / M48Z35Y-70PC1 NVRAM (28-pin, same JEDEC pinout as a 62256-family 32Kx8 SRAM).
The symbol uses a LOGICAL pin layout (A0–A14/GND left, VCC/Q0–Q7/CE–OE–WE right) instead of the physical DIP pin order, for readability -- pin NUMBERS still match the real chip exactly. It occupies 0000h–7FFFh (32KB) and simulates a read-only EPROM for code storage:
- /CE (pin 20) <- NVRAM_CS_n from the SPLD: asserts only when A15=0 AND PSEN_n=0 (an instruction fetch in the lower 32K).
- /OE (pin 22) <- /PSEN directly, NOT gated by address decode, so the part only ever drives the bus on a program-memory read, never a data read.
- /WE (pin 27) tied permanently to VCC -- writes are never performed to this chip in this lab (not used as read/write SRAM until a later lab).
- The board's second 28-pin socket (silkscreened 'SRAM') is left EMPTY per the lab handout -- do not solder a second socket there this lab.
- Fitted in a 28-pin wire-wrap DIP socket, NOT the ZIF socket from the tool kit (returned with the kit at end of semester).

Reference: ECEN 5613 PCB partial schematic (c) 2005 Linden H. McClure, Ph.D.
Drawn by: Nathan Cohen
NVRAM (EPROM substitute) and LED driver
Power supply, reset, oscillator, 8051, serial connector, address latch, SPLD decode,
University of Colorado Boulder – ECEN 5613

Sheet: /
File: ECEN5613_Dev_Board.kicad_sch

Title: ECEN 5613 Embedded System Design – Development Board

Size: A2	Date: 2026-09-24	**Rev: C**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1